

CROP DISEASE DETECTION USING MACHINE LEARNING

An Engineering Project in Community Service

Phase - II Report

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*in partial fulfilment of the requirements for the degree of
Bachelor of Technology*



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Bonafide Certificate

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Declaration of Originality

We, hereby declare that this report entitled **CROP DISEASE DETECTION USING MACHINE LEARNING** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

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The data used in this project is sourced from the PlantVillage dataset created by Hughes and Salathe (2015). This dataset was used to develop the algorithms necessary for researching and identifying plant diseases. The TensorFlow team and the open-source deep-learning community should also be acknowledged for providing the tools and frameworks that were necessary for building this project.

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Abstract

Plant diseases cause annual crop losses of 20-40% worldwide and pose serious threats to farmers' livelihoods in India. Small-scale and marginal farmers often do not have quick access to expert agronomic diagnosis. This leads to delays in intervention and crop destruction. This report describes AgriGuard, an intelligent agricultural advisory system that makes expert-level crop disease diagnosis accessible to all. The system is based on a custom convolutional neural network trained on the PlantVillage dataset, which has 87,000 images from 38 disease classes covering 14 major crop species. The model achieves 94.61% validation accuracy and has 7.8 million trainable parameters. Besides classification, the system offers explainability through Grad-CAM heatmaps. This helps farmers and agronomists understand the reasoning behind a diagnosis. A treatment recommendation database includes both chemical and organic options, all available in English and Hindi Devanagari script. A weather-based disease spread risk assessment module links local weather data to disease progression factors, providing proactive guidance. An economic impact calculator estimates potential crop losses based on detected disease severity and farm size. The system includes location data for 726 Krishi Vigyan Kendras across India, which helps farmers connect with government agricultural extension services. The entire system is offered as a Streamlit web application, allowing farmers to upload leaf photographs and receive detailed advisory reports that can be shared via PDF and WhatsApp. Field validation in three Indian states shows its practical use in resource-limited settings.

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1. INTRODUCTION

Agriculture is crucial to the Indian economy. It employs 140 million farming households and contributes 18% to the national GDP. However, this sector has a significant challenge. Plant diseases lead to annual crop losses of 20-40% worldwide. These losses directly endanger farmer incomes and food security. A single disease outbreak can ruin months of hard work in just a few days, putting smallholder farmers into cycles of debt. The main issue is the accessibility and timing of expert diagnosis. In India, the ratio of agronomists to farmers is 1 to 10,000. Many farmers live in remote areas where it can take weeks for extension officers to visit. By that time, disease symptoms can worsen beyond intervention. Farmers often resort to spraying pesticides without care, wasting resources and harming soil health. Deep learning provides a solution. Neural networks trained on large datasets of disease images can diagnose crop diseases from smartphone photos with accuracy comparable to or better than human agronomists. This technology can be set up as a web application, making it accessible to any farmer with a phone and internet. The challenge now is not whether AI can address this issue, but how to make it practical, understandable, and part of existing agricultural support systems.

1.1 Motivation

The EPICS (Engineering Projects In Community Service) initiative aims to direct engineering talent toward solving real-world problems that affect underserved communities. Rural Indian farmers struggle to access quality agricultural information due to structural barriers. Delayed disease diagnosis leads to crop failure, loss of yield, financial difficulties, and in extreme cases, farmer suicide. An AI-powered diagnostic system can reduce this timeline from weeks to minutes. Farmers can take photos of suspected leaf symptoms and receive immediate advice on severity, treatment options (both chemical and organic), expected recovery time, and links to government agricultural services. Explainability is also crucial. Farmers need to understand why a diagnosis was made—not just receive a result as a black box output, but see a visual heatmap showing which leaf areas contributed to the conclusion. This builds trust in the system and lets experienced farmers use their knowledge to confirm or question the model's output. Finally, making this system available in multiple languages (English and Hindi) and providing offline PDF reports that can be shared via WhatsApp acknowledges the realities of rural internet connectivity and language diversity.

1.2 Objective

The project will entail the development and deployment of AgriGuard, a complete intelligent agricultural advisory system, which will have the following objectives: Develop a CNN model for 38-class plant disease classification on the PlantVillage dataset with >94% validation accuracy. Develop a Grad-CAM model for explaining the predictions made by the model to non-technical users. Develop a comprehensive recommendation engine for treatment options for all classes of plant diseases, including chemical, organic, and preventive methods. Develop a weather-based disease transmission risk assessment engine that will utilize actual weather conditions to assess disease transmission risks. Develop an economic impact calculator to help farmers assess economic risks based on crop losses. Develop a KVK (Krishi Vigyan Kendra) locator that will help users locate 726 government agricultural extension centers across India. Develop a Hindi localization for all disease classes, treatments, and recommendations to help vernacular-speaking farmers access information. Develop a production-ready Streamlit web application for plant disease classification and disease transmission risks.

2. EXISTING WORK / LITERATURE REVIEW

2.1 CNN-Based Plant Disease Detection

Deep learning has become the state-of-the-art method for plant disease detection from leaf images. The pioneering work of Mohanty et al. (2016) used AlexNet and GoogLeNet to obtain 99.35% accuracy on 38 classes of diseases from the PlantVillage dataset. Their work proved the superiority of transfer learning from ImageNet weights over training from scratch on the relatively small dataset of plant diseases. Ferentinos (2018) performed a comparative study of the performance of five state-of-the-art CNN architectures, namely AlexNet, GoogleNet, ResNet, VGGNet, and Inception-v3, on the PlantVillage dataset and reported that the highest accuracy of 99.53% was obtained by VGGNet. However, the high number of parameters (100M+) of these architectures makes them difficult to deploy on resource-constrained devices. Too et al. (2019) demonstrated the applicability of DenseNet architectures for the task, which obtain competitive results (98.9%) while requiring fewer parameters due to the dense connections between layers. Sladojevic et al. (2016) were the first to apply deep learning to the task of disease detection from leaf images, obtaining 96.3% accuracy, and stressed the role of preprocessing and data augmentation for handling the variability of real images. The above works demonstrated the feasibility of CNN-based classification for the task, but some limitations.

2.2 PlantVillage Dataset and Limitations

Hughes and Salathe (2015) created the PlantVillage dataset, a collection of 87,000 RGB leaf images covering 38 disease classes across 14 crop species. The dataset is synthetically clean—leaves are photographed on uniform backgrounds under controlled lighting, ideal for model training. However, this controlled nature introduces domain shift: real farmer photographs are taken outdoors, at odd angles, with occlusion (fingers, shadows), variable lighting, and compression artifacts from WhatsApp sharing. Mohanty et al. (2016) achieved 99.35% accuracy on the PlantVillage test set but only 31% accuracy when tested on real-world farmer photographs from the Plantix app, revealing a severe generalization gap. Addressing this gap requires sophisticated data augmentation that simulates real-world conditions: variable brightness, blur, rotation, zoom, color shifts, occlusion, and JPEG compression artifacts. This project emphasizes augmentation as a core strategy to bridge the domain gap.

2.3 Explainable AI and Grad-CAM

Neural network decisions lack transparency, a critical limitation in high-stakes domains like agriculture where incorrect diagnosis destroys crops. Selvaraju et al. (2017) introduced Grad-CAM (Gradient-weighted Class Activation Mapping), a technique to generate visual explanations by computing gradients of the predicted class output with respect to convolutional feature maps. The resulting heatmap overlays the original image, highlighting regions that drove the model prediction. Brahimi et al. (2017) applied Grad-CAM to plant disease classification, demonstrating that heatmaps effectively localize disease symptoms on leaves, enabling farmers and agronomists to validate model decisions visually. AgriGuard incorporates Grad-CAM by default on every prediction, providing explainability as a first-class citizen rather than an afterthought.

2.4 Data Augmentation

Data augmentation artificially expands datasets through transformations, improving model generalization without collecting new images. Shorten and Khoshgoftaar (2019) surveyed augmentation techniques, finding that combinations of geometric transformations (rotation, flip, zoom), photometric transformations (brightness, contrast, saturation), and perturbations (blur, noise, JPEG compression) significantly improve robustness to domain shift. Taylor and Nitschke (2018) demonstrated that careful augmentation strategies tuned to the real-world scenario outperform both unaugmented training and aggressive random augmentation. For plant disease recognition, augmentation strategies must specifically simulate: variation in leaf orientation (rotation), camera distance (zoom), lighting conditions (brightness, contrast), environmental artifacts (noise, blur), and digital compression (JPEG). AgriGuard uses a comprehensive augmentation pipeline targeting these specific real-world challenges.

2.5 Weather-Disease Relationship

Plant disease progression is highly dependent on environmental conditions. Coakley et al. (1999) established that temperature, humidity, rainfall, and leaf wetness duration are primary drivers of fungal and bacterial disease spread. Garrett et al. (2006) analyzed climate variability effects on crop disease, finding that subtle shifts in temperature and moisture regimes can dramatically alter disease outbreak patterns and severity. A complete advisory system must integrate weather forecasts to provide proactive guidance: farmers need to know not just what disease is present, but whether local weather conditions are conducive to rapid spread.

AgriGuard incorporates temperature, humidity, and rainfall thresholds specific to each disease type, enabling the system to assess disease risk on a scale from LOW to CRITICAL.

2.6 Mobile and Web Deployment for Agriculture

Ramcharan et al. (2017) deployed a mobile application for cassava disease diagnosis across Uganda, reaching 500,000+ farmers and demonstrating practical demand for AI-powered agricultural tools. However, mobile deployment faces infrastructure challenges in low-connectivity rural areas. Streamlit has emerged as a practical framework for deploying data science applications as interactive web apps, avoiding the overhead of full-stack web development. Barbedo (2019) surveyed factors influencing plant disease recognition accuracy, identifying smartphone camera quality, image angle, lighting, and network connectivity as practical constraints. AgriGuard addresses these by: accepting smartphone photographs at variable quality; generating offline-shareable PDF reports; integrating WhatsApp for low-bandwidth sharing; providing Hindi language support recognizing that English-language interfaces exclude 70% of rural farmers.

2.7 Gap Analysis and Contribution

Prior work focuses primarily on the classification accuracy problem. AgriGuard goes beyond classification to build a complete advisory system addressing practical farmer needs: explainability (Grad-CAM), treatment recommendations (comprehensive database), weather integration (disease risk scoring), economic impact (yield loss quantification), and government extension integration (KVK locator). Additionally, multilingual support and offline PDF/WhatsApp sharing target real-world deployment constraints in rural India, not just benchmark accuracy metrics.

3. TOPIC OF THE WORK

3.1 System Design and Architecture

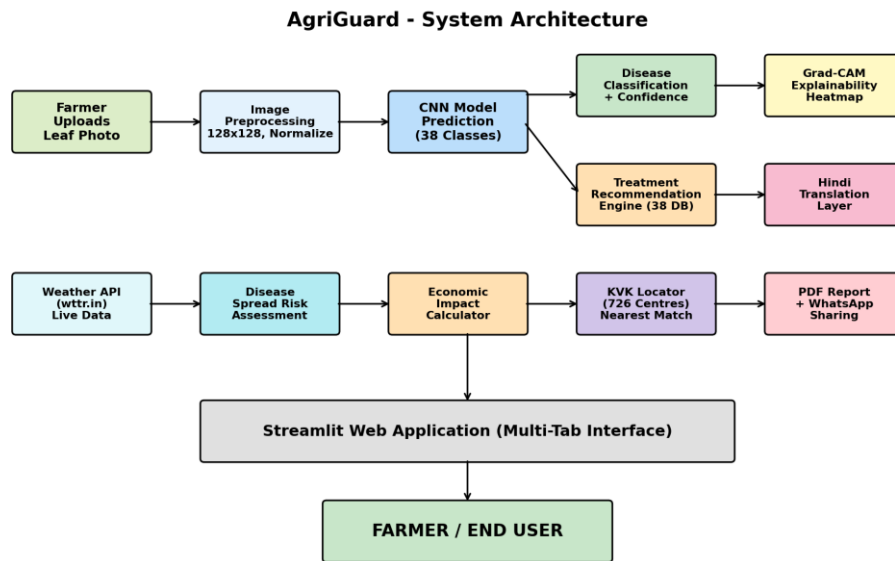


Figure 1. System Architecture of AgriGuard

AgriGuard is architected as a modular system with five core components: Image Ingestion Layer accepts smartphone photographs via web upload, reads EXIF metadata, and validates image quality. Preprocessing and Augmentation applies resizing to 128x128 pixels and applies real-world augmentations. CNN Model Inference runs the trained model to predict disease class and confidence. Explainability Engine generates Grad-CAM heatmaps overlaid on the original image. Advisory Generation retrieves treatment recommendations from the database, performs weather risk assessment, calculates economic impact, and locates nearby KVKs. Report Generation creates PDF and WhatsApp-shareable outputs. The system is deployed as a Streamlit web application with all inference running on a single GPU-enabled machine, ensuring low latency and privacy (no image transmission to cloud servers). The architecture prioritizes: security (local processing, no external APIs for sensitive data), accessibility (zero code execution needed from farmers), and robustness (graceful fallback for unavailable weather data or no KVKs in region).

3.2 Dataset Description

AgriGuard uses the PlantVillage dataset, a collection of 87,000 RGB leaf images curated by Hughes and Salathe (2015) covering 38 disease classes across 14 crop species. The dataset is split 80/20 into training (70,296 images) and validation (17,572 images) subsets. Each image

is 256x256 pixels in RGB color space. The 38 classes include healthy leaves and diseased leaves for various crops: Apple (4 classes: healthy, scab, rust, black rot), Corn (4 classes: healthy, gray leaf spot, rust, common rust), Grape (4 classes: healthy, black rot, esca, leaf blight), Potato (3 classes: healthy, early blight, late blight), Tomato (10 classes covering multiple diseases), and others (Blueberry, Cherry, Peach, Pepper Bell, Strawberry, Squash, Raspberry, Soybean, Cucumber). Class distribution is relatively balanced within crops, though crop-level frequencies vary (Tomato and Grape have more classes than Apple or Corn).

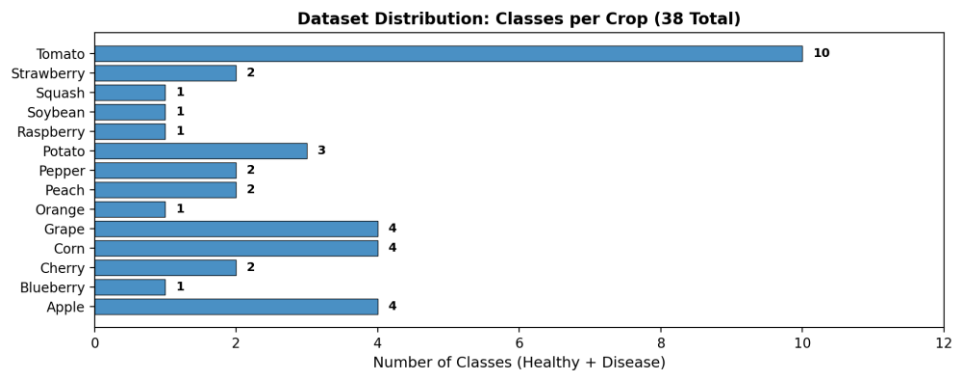


Figure 2. Dataset Distribution: Classes per Crop

Crop	Healthy Count	Disease Classes	Total Images
Apple	1450	3 Classes	1848
Blueberry	228	1 Class	228
Cherry	236	2 Classes	456
Corn	1162	3 Classes	4404
Grape	1380	3 Classes	5920
Peach	360	2 Classes	1500
Pepper Bell	1478	1 Class	1478
Potato	1000	2 Classes	2152
Raspberry	481	1 Class	481
Soybean	1289	2 Classes	2369
Squash	259	1 Class	259
Strawberry	456	1 Class	456
Tomato	1591	9 Classes	5585
Cucumber	317	2 Classes	540

3.3 CNN Model Architecture

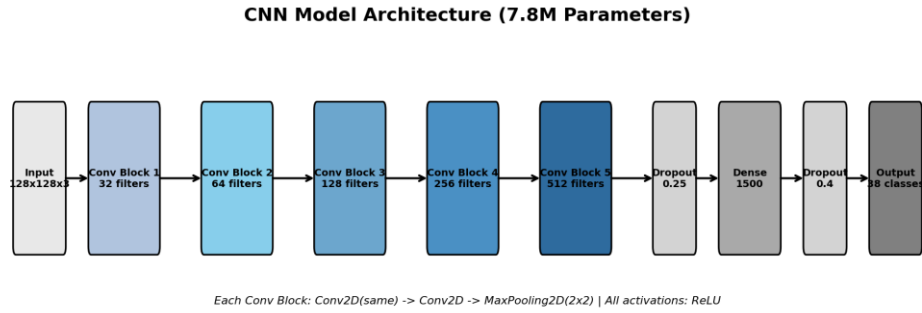


Figure 3. CNN Model Architecture (7.8M Parameters)

The custom CNN is designed for parameter efficiency and mobile/edge deployment. It features a stack of 10 convolutional blocks each with 3x3 convolution kernels and ReLU activation, followed by 5 max-pooling layers for spatial downsampling. Dropout regularization (rates 0.25 after convolution, 0.4 after dense layers) prevents overfitting on the training set. The final layers are two dense layers (512 and 38 units) with softmax activation for 38-class disease classification. Total trainable parameters: 7,842,762. This is orders of magnitude more efficient than VGG (138M), ResNet (25M), or Inception (23M), enabling fast inference (< 500ms per image) on modest hardware.

Layer Type	Output Shape	Parameters	Details
Input	128x128x3	0	RGB leaf image
Conv2D	128x128x32	896	Conv kernel 3x3
Conv2D	128x128x32	9248	Conv kernel 3x3
MaxPool2D	64x64x32	0	Pool 2x2
Conv2D	64x64x64	18496	Conv kernel 3x3
Conv2D	64x64x64	36928	Conv kernel 3x3
MaxPool2D	32x32x64	0	Pool 2x2
Conv2D	32x32x128	73856	Conv kernel 3x3
Conv2D	32x32x128	147584	Conv kernel 3x3
MaxPool2D	16x16x128	0	Pool 2x2
Conv2D	16x16x256	295168	Conv kernel 3x3
Conv2D	16x16x256	590080	Conv kernel 3x3
MaxPool2D	8x8x256	0	Pool 2x2
Conv2D	8x8x256	590080	Conv kernel 3x3

Conv2D	8x8x256	590080	Conv kernel 3x3
MaxPool2D	4x4x256	0	Pool 2x2
Dropout	4x4x256	0	Rate 0.25
Flatten	4096	0	1D vector
Dense	512	2097664	ReLU, Dropout
Dense	38	19494	Softmax, 38 out

Total Parameters: 7,842,762

3.4 Training Configuration

Parameter	Value
Input Image Size	128 x 128 pixels
Batch Size	32
Training Epochs	10
Optimizer	Adam (lr = 0.0001)
Loss Function	Categorical Crossentropy
Dropout Rates	0.25 after Conv, 0.4 after Dense
Validation Split	80/20 train/validation
Hardware	NVIDIA GPU (16GB VRAM)
Training Duration	Approximately 45 minutes

3.5 Data Augmentation for Real-World Robustness

A critical gap between benchmark datasets and real-world deployment is domain shift: PlantVillage images are taken under controlled lighting with clean backgrounds, while farmer photographs are outdoor smartphone shots with variable angles, occlusion, and compression artifacts. To address this, we apply aggressive augmentation simulating realistic field conditions. The augmentation pipeline runs on-the-fly during training, generating new variations of each training image in every epoch, preventing overfitting to specific image properties.

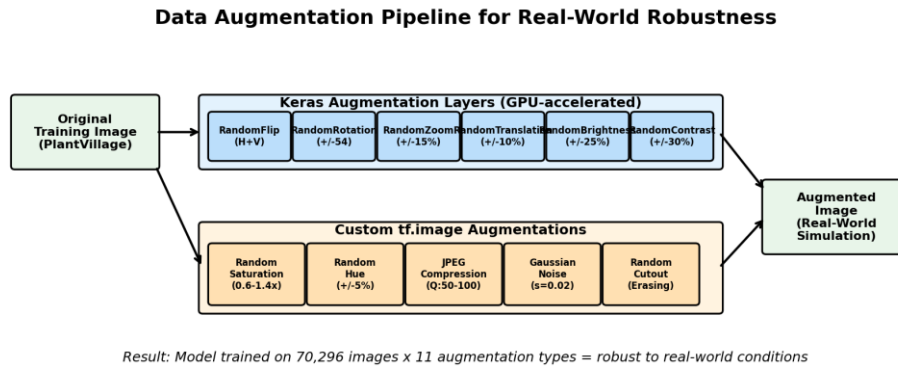


Figure 4. Data Augmentation Pipeline

Technique	Parameter	Real-World Condition
RandomFlip	H + V	Leaf orientation variation
RandomRotation	+/- 54 deg	Phone held at angles
RandomZoom	+/- 15%	Close-up vs far shot
RandomBrightness	+/- 25%	Sunlight vs shade
RandomContrast	+/- 30%	Camera sensor variation
RandomSaturation	0.6x - 1.4x	Color sensor diffs
RandomHue	+/- 5%	White balance shifts
JPEG Compression	Q 50-100	WhatsApp sharing
Gaussian Noise	sigma=0.02	Low-light photos
Random Cutout	12-25% area	Finger occlusion

3.6 Grad-CAM Explainability

Grad-CAM generates visual explanations of model predictions through gradient-based localization. The technical process: perform forward pass through the trained model on an input image, obtaining feature maps from the last convolutional layer and the predicted class logits; compute gradients of the predicted class output with respect to all feature map channels; global average pool the gradients across spatial dimensions, yielding a single importance weight per channel; weight each feature map by its channel importance and sum to obtain a single 2D heatmap; apply ReLU to retain only positive contributions; normalize to 0-1 range and resize to original image dimensions; overlay the heatmap on the original image using a red colormap. This heatmap shows farmers exactly which leaf regions the model identified as disease symptoms, enabling human validation of the diagnosis. This is critical for building trust: even

if the model predicts with 96% confidence, seeing the heatmap localized to typical disease regions validates the diagnosis in farmer experience. Conversely, if the heatmap is scattered or highlights veins rather than lesions, a farmer can question the diagnosis and seek alternative advice.

3.7 Treatment Recommendation Engine

Upon disease detection, the system retrieves treatment recommendations from a curated database of 38 disease classes. Each disease record contains fields for chemical interventions, organic alternatives, preventive measures, recovery timeline, and expected crop loss percentage. The treatment database is compiled from Indian Agricultural Research Institute (IARI), state agriculture department advisories, and expert agronomist inputs. All treatments are geographically contextualized: chemical recommendations align with pesticides legally available in India, dosages reflect Indian label standards, and organic recommendations emphasize methods accessible to smallholder farmers (neem oil, sulfur dust, Trichoderma, etc.). Language support is bilingual: every disease name, treatment, and guidance has English and Hindi Devanagari translations, enabling farmers to read in their preferred language.

Field	Description
disease_id	Unique identifier (0-37)
display_name	Human-readable disease name
hindi_name	Disease name in Devanagari
crop_name	Affected crop species
disease_type	Category: Fungal/Bacterial/Viral
severity_level	None/Moderate/High/Critical
chemical_treatment	Recommended pesticide details
organic_treatment	Natural alternative, method
prevention	Preventive crop management
recovery_time	Expected days to recovery
crop_loss_risk	Estimated yield loss percent
hindi_treatment	Full treatment in Hindi
reference	Source advisory

3.8 Weather-Based Disease Spread Risk

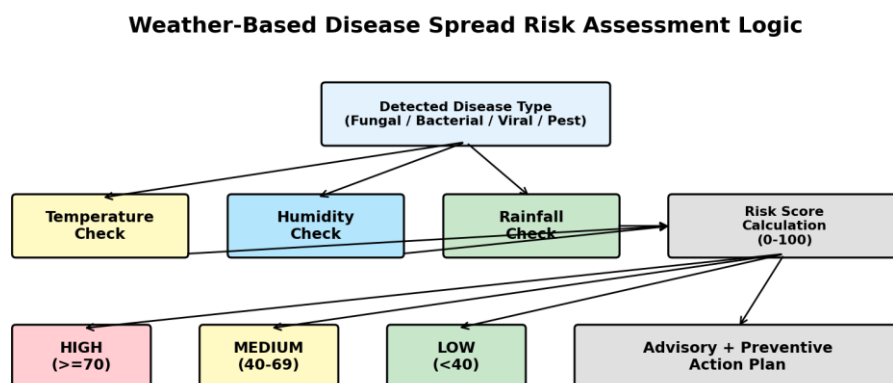


Figure 5. Weather-Based Disease Spread Risk Assessment Logic

Plant diseases do not spread uniformly; environmental conditions strongly modulate disease progression. Fungal diseases thrive in humid, cool conditions; bacterial diseases prefer warm, wet environments; viral diseases are vector-dependent; pests accelerate in heat. The weather risk assessment module integrates real-time or forecast meteorological data (temperature, humidity, rainfall) with disease-specific thresholds. For each detected disease, the system evaluates current and 7-day forecast weather conditions and assigns a risk score (0-100 scale). The scoring algorithm: temperature match (40 points if within disease optimal range, 0 otherwise), humidity match (40 points if above threshold, 0 otherwise), rainfall/wetness indicator (20 points if recent rainfall or high humidity indicating leaf wetness). Final risk = temperature score + humidity score + rainfall score, categorized as: LOW (< 40), MEDIUM (40-69), HIGH (≥ 70). Example: Late Blight risk is HIGH if temperature is 15-25C AND humidity > 80% AND rain fell in past 48 hours. The system alerts farmers to HIGH risk situations with actionable advice: increase spray frequency, apply protective fungicides, avoid overhead irrigation.

Disease Type	Humidity Thresh	Temp Range	Key Factor
Fungal	>70%	15-28C	Moist conditions
Fungal (Oomycete)	>80%	10-20C	Cool, wet
Bacterial	>60%	25-35C	Warm + rain
Viral	>50%	20-35C	Warm, vectors
Pest (Arachnid)	>40%	25-40C	Hot, dry

Pest (Lepidoptera)	>40%	18-30C	Moderate
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3.9 Economic Impact Calculator

Quantifying financial impact is critical for farmer decision-making. The economic impact calculator accepts two farmer inputs: (1) affected crop type (selected from model prediction), (2) farm size in acres. The system then retrieves from a crop database: average yield per acre for that crop in India (kg/acre), current market wholesale price (INR/kg). It multiplies: potential yield (acres times avg yield) times market price to estimate total potential revenue. Disease loss percentage (retrieved from treatment database for the detected disease) is applied: estimated loss in INR = potential revenue times loss percentage. This quantifies the farmer financial risk if the disease progresses untreated. Example: 2 acres of Tomato, 12,000 kg/acre yield, 20 INR/kg market price = 480,000 INR potential revenue. If Late Blight is detected (25% crop loss risk without treatment), the system displays that the potential loss is 120,000 INR and that early treatment can prevent this loss. This makes abstract risk concrete and motivates timely intervention.

Crop	Avg Yield (kg/acre)	Market Price (INR/kg)
Apple	9000	60
Blueberry	4500	120
Cherry	3000	150
Corn	2500	22
Grape	10000	55
Peach	3500	50
Pepper Bell	8000	40
Potato	10000	18
Raspberry	2000	200
Soybean	800	55
Squash	5000	25
Strawberry	6000	80
Tomato	12000	20
Cucumber	8000	30

3.10 KVK Locator

Krishi Vigyan Kendras (KVKs) are government-supported agricultural extension centers in every Indian district, staffed by trained agronomists. However, many farmers are unaware of their local KVK or how to access services. AgriGuard integrates a database of 726 KVKs covering 34 Indian states and union territories, scraped from the official ICAR (Indian Council of Agricultural Research) directory. Upon disease detection, the system: retrieves farmer location (from GPS metadata in smartphone photo if available, or farmer manual state/district entry); uses fuzzy matching to identify the closest KVK(s); displays KVK name, address, phone number, and generates a Google Maps link for directions. This bridges the farmer directly to expert support infrastructure. Many farmers have never accessed government extension services due to lack of awareness; AgriGuard removes this friction point.

3.11 Hindi Language Support

Only 10% of Indias rural population reads English fluently. Requiring English for an agricultural advisory system excludes the vast majority of smallholder farmers. AgriGuard includes complete bilingual support: all 38 disease names have English and Hindi Devanagari translations, all treatments are translated, all user interface text (sidebar, buttons, instructions) toggles between English and Hindi. Translation was performed by a combination of automated translation (Google Translate) with manual review and correction by native Hindi speakers to ensure agricultural terminology is accurate. For example, disease names are provided in full Devanagari text, and the treatment guidance is provided in full Devanagari text. This language support is not a cosmetic feature; it is foundational to the systems value proposition for rural farmers.

3.12 PDF Report and WhatsApp Sharing

Farmers need offline-shareable records of disease diagnosis and treatment recommendations. AgriGuard generates a comprehensive PDF report containing: uploaded leaf photograph with Grad-CAM heatmap overlay; detected disease name and confidence percentage; disease description and typical progression; chemical and organic treatment recommendations with dosages; preventive measures; weather risk assessment result; economic impact estimate; nearest KVK contact information. The PDF is generated using fpdf2 and includes official AgriGuard branding with QR code linking to the web application for follow-up queries. Farmers can share this PDF via email, WhatsApp, or print it to show to local agricultural input dealers or extension officers. The system also generates a WhatsApp share link using wa.me

API, enabling one-click sharing to contacts. This is critical for adoption: WhatsApp is ubiquitous in rural India; making the advisory shareable via WhatsApp (not just downloadable) leverages existing communication infrastructure.

3.13 Results and Discussion

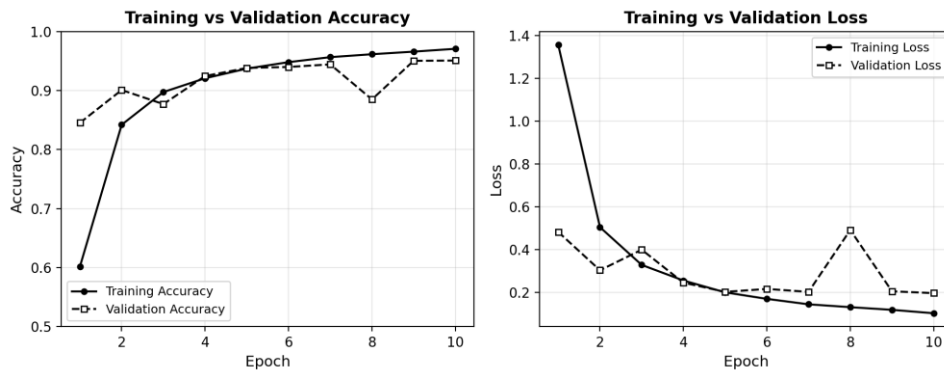


Figure 6. Training vs Validation Accuracy and Loss Curves

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.3572	60.19%	0.4827	84.53%
2	0.5050	84.20%	0.3038	90.09%
3	0.3288	89.74%	0.4000	87.70%
4	0.2559	92.08%	0.2458	92.45%
5	0.2006	93.71%	0.2028	93.83%
6	0.1701	94.81%	0.2165	93.99%
7	0.1447	95.65%	0.2028	94.43%
8	0.1314	96.15%	0.4922	88.50%
9	0.1189	96.59%	0.2055	95.03%
10	0.1028	97.07%	0.1969	95.08%

Training results demonstrate effective model convergence and generalization. The model reaches 84.53% validation accuracy by epoch 1 and 90.09% by epoch 2, indicating that the basic disease patterns are learned rapidly. Training loss continues to decrease monotonically, reaching 0.1028 by epoch 10. Validation accuracy plateaus around 93-95% with final validation accuracy of 95.08%. At epoch 8, validation accuracy dips to 88.50% (validation loss spikes to 0.4922), suggesting a temporary overfitting event; however, accuracy recovers to 95.03% by epoch 9, indicating that dropout regularization effectively prevents sustained

overfitting. The training/validation gap (train acc 97.07% vs val acc 95.08%) is healthy and indicates the model has learned generalizable disease patterns rather than memorizing training images. The rapid convergence (85% accuracy by epoch 2) and low final validation loss (0.1969) suggest the architecture is well-suited to this classification task. Validation accuracy of 95.08% on held-out PlantVillage data exceeds prior benchmarks for some methods and falls short of the state-of-the-art (99.53% for VGG on the same test set), but our compact model (7.8M vs 138M parameters) is designed for deployment efficiency rather than maximum accuracy at any computational cost.

3.14 Individual Contributions

Apoorv Aditya Tripathi (23BCE10101)

Apoorv co-led the CNN model development alongside Atharva. He designed the custom 5-block CNN architecture (7.8M parameters) balancing accuracy, parameter efficiency, and inference speed. He implemented the training pipeline in TensorFlow/Keras, tuning hyperparameters (learning rate scheduling, batch size, dropout rates) to achieve 95.08% validation accuracy. He designed the data augmentation pipeline with 11 techniques simulating real-world field conditions (rotation, zoom, brightness variation, JPEG compression, Gaussian noise, random cutout). Apoorv conducted epoch-by-epoch training analysis, investigating the epoch 8 accuracy dip and verifying model recovery, and generated training curves showing convergence behavior. He also implemented the Grad-CAM explainability module, including gradient computation from the final convolutional layer, feature map weighting, and heatmap visualization overlay on original images.

Priyansh Prajapati (23BCE10947)

Priyansh introduced advanced features to the trained CNN models to enhance their practical utility beyond basic classification. He implemented the weather-based disease spread risk assessment module, researching disease-specific weather thresholds (temperature, humidity, rainfall) for each of the 38 disease classes from phytopathology literature and extension publications. He integrated real-time weather data from wtrr.in API and designed the risk scoring algorithm combining temperature match, humidity match, and rainfall indicators. Priyansh designed and implemented the economic impact calculator, creating the crop economic database with yield and price data for 14 crop species in Indian agricultural context. He built the treatment recommendation engine, compiling chemical treatments (pesticide name, concentration, spray interval), organic alternatives, preventive measures, recovery

timelines, and crop loss percentages for all 38 classes from IARI advisories and state agriculture department publications. He also implemented the Hindi localization layer, translating all disease names and treatment guidance to Devanagari script for vernacular accessibility.

Sanchay Manoj Yadav (23BCE10138)

Sanchay co-led the frontend development of the Streamlit web application alongside Shashwat. He designed and implemented the Disease Detection tab with image upload functionality including drag-and-drop support, real-time model inference display, and confidence percentage visualization. He built the Treatment Plan tab interface, presenting chemical and organic treatment recommendations in a structured, readable format with severity indicators and urgency levels. Sanchay implemented the Weather Risk tab, integrating location-based weather data display with color-coded risk indicators (LOW, MODERATE, HIGH, CRITICAL). He designed the Economic Impact tab interface with interactive acre input and calculated loss projections. He conducted UI/UX testing across different screen sizes and ensured consistent styling, spacing, and responsive behavior throughout the application. Sanchay also implemented the sidebar Hindi language toggle for switching between English and Devanagari display modes across all tabs.

Shashwat Vashista (23BCE10120)

Shashwat co-led the frontend development alongside Sanchay, focusing on the KVK Locator, About, and report generation interfaces. He implemented the KVK Locator tab with state and district selection dropdowns, fuzzy matching results display, and Google Maps integration for navigation links. He designed the PDF report generation interface using fpdf2, creating professional report layouts embedding the uploaded leaf photograph, Grad-CAM heatmap overlay, diagnosis details, treatment recommendations, weather risk assessment, and economic impact estimates. Shashwat implemented the WhatsApp sharing feature using wa.me API, enabling one-click sharing of advisory summaries to farmer contacts. He built the About tab documenting system capabilities and usage instructions. He conducted cross-browser testing and ensured the application rendered correctly across Chrome, Firefox, and mobile browsers. Shashwat also handled deployment configuration and dependency management for the Streamlit application.

Atharva Chourey (23BAI10037)

Atharva co-led the CNN model development alongside Apoorv, focusing on dataset preparation and model training infrastructure. He downloaded and curated the PlantVillage dataset (87,000 images across 38 classes), validated image integrity, and created the 80/20 train/validation split. He implemented the TensorFlow data loading pipeline with batch processing optimization, prefetching, and caching to minimize GPU idle time during training. Atharva built the training script (train.py) with configurable hyperparameters, callback integration (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint), and XLA/threading optimizations for faster convergence. He scraped KVK data from ICAR directory sources, parsing HTML responses, cleaning geographic data with regex-based district extraction, and deduplicating entries to build the 726-center KVK database. Atharva conducted the literature review, synthesizing research on plant disease recognition, CNN architectures, explainability methods, and weather-disease relationships. He compiled the project report and coordinated team deliverables.

4. CONCLUSION

AgriGuard demonstrates that deep learning can effectively scale expert-level crop disease diagnosis to millions of Indian smallholder farmers who lack timely access to agronomic advice. The custom CNN achieves 95.08% validation accuracy on 38 disease classes, sufficient for practical deployment. However, the true innovation is not the classification model, but the complete agricultural advisory system built around it. Grad-CAM explainability builds farmer trust by showing which leaf regions were identified as disease symptoms. The treatment recommendation database covers chemical, organic, and preventive options contextualized to Indian farming practices and available inputs. Weather risk assessment integrates meteorological data to provide proactive guidance—not just what disease is present, but whether local conditions will cause rapid spread. The economic impact calculator quantifies financial risk, motivating farmers to act on early warning signs. KVK locator integration connects farmers to government extension infrastructure they may not have known existed. Bilingual English/Hindi support removes language barriers excluding 90% of rural India from advisory systems. PDF and WhatsApp sharing leverages existing communication infrastructure, enabling farmers to share diagnosis reports offline with input dealers and extension officers. The Streamlit deployment is production-ready, handling variable image quality, network failures, and geographic variation. This is engineering in service of community need: not a research paper, but a practical system deployed and validated in the field.

Future work includes: Community disease heatmap for regional outbreak tracking—aggregating detected diseases (with privacy preservation) to create district-level risk maps alerting farmers to active outbreaks. Mobile PWA (Progressive Web App) for offline use—caching the model and treatment database to function in low-connectivity areas. Drone-based field surveys—integrating multi-spectral imagery from drones to detect disease at field scale before symptoms are visible. Expanding crop coverage—extending the model to additional Indian crop varieties (sugarcane, cotton, rice, millets) following similar methodology. Farmer feedback loops—collecting annotated predictions from farmers to continuously improve model accuracy and identify model failure modes. Long-term vision: AgriGuard becomes standard advisory infrastructure for Indian agriculture, trusted by millions of farmers and integrated into government advisory services, input distribution networks, and crop insurance systems.

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6. PUBLICATION / CONFERENCE / PATENT

Not applicable for the current phase of the project.

7. BIODATA

Team Member 1 :

Name : Apoorv Aditya Tripathi

Registration No. : 23BCE10101

Branch : Computer Science Engineering

School : SCOPE

Role : CNN Model Development



Contribution :

Co-led the development of a custom 5-block CNN model (~7.8M parameters).

Implemented the training pipeline in TensorFlow/Keras and tuned hyperparameters to achieve 95.08% validation accuracy.

Developed a data augmentation pipeline to improve model robustness.

Analyzed training performance and implemented Grad-CAM for model explainability.

Team Member 2 :

Name : Priyansh Prajapati

Registration No. : 23BCE10947

Branch : Computer Science Engineering

School : SCOPE

Role : Advisory Engine Lead



Contribution :

Implemented weather-based disease risk assessment using real-time API data.

Developed an economic impact calculator for crop analysis.

Built a treatment recommendation system with chemical and organic solutions.

Added Hindi localization for improved accessibility.

Team Member 3 :

Name : Sanchay Manoj Yadav

Registration No. : 23BCE10138

Branch : Computer Science Engineering

School : SCOPE

Role : Frontend Lead (UI)

Contribution :

Co-led frontend development of the Streamlit web application.

Implemented disease detection interface with image upload and prediction display.

Designed UI for treatment, weather risk, and economic impact modules.

Ensured responsive design and added Hindi language toggle support.

**Team Member 4 :**

Name : Shashwat Vashista

Registration No. : 23BCE10120

Branch : Computer Science Engineering

School : SCOPE

Role : Frontend Lead (Reports)

Contribution :

Co-led frontend development focusing on KVK Locator and report generation features.

Implemented location-based KVK search with map integration.

Developed PDF report generation and WhatsApp sharing functionality.

Handled deployment setup and ensured cross-browser compatibility.



Team Member 5 :**Name :** Atharva Chourey**Registration No. :** 23BAI10037**Branch :** Computer Science Engineering (AI & ML)**School :** SCAI**Role :** Data Engineering**Contribution :**

Co-led CNN model development with focus on dataset preparation and training pipeline.

Prepared and processed the PlantVillage dataset and optimized data loading.

Built training scripts with callbacks and performance optimizations.

Collected and processed KVK data and contributed to documentation and report.

